

100 MCQs on Data Structures and Algorithms (DSA) with Answers



Arrays and Strings (15 MCQs)

- What is the time complexity to access an element in an array by index?
 - ☐ a) $O(1)$ ✓
 - ☐ b) $O(n)$
 - ☐ c) $O(\log n)$
 - ☐ d) $O(n \log n)$
- What will be the output of: "abc".substring(1,3) in Java?
 - ☐ a) ab
 - ☐ b) bc ✓
 - ☐ c) ac
 - ☐ d) abc
- Which method is used to find the length of an array in Java?
 - ☐ a) length()
 - ☐ b) size()
 - ☐ c) length ✓
 - ☐ d) count()
- Which data structure is used to implement strings in most programming languages?
 - ☐ a) Array ✓
 - ☐ b) Stack
 - ☐ c) Queue
 - ☐ d) Graph
- What is the output of: "abc" + 1 + 2?
 - ☐ a) abc3
 - ☐ b) abc12 ✓
 - ☐ c) 3abc
 - ☐ d) 1abc2
- How is a 2D array represented in memory?
 - ☐ a) As an array of arrays ✓
 - ☐ b) As a tree
 - ☐ c) As a stack
 - ☐ d) As a graph
- Which of the following is not a valid string operation?
 - ☐ a) Concatenation
 - ☐ b) Slicing
 - ☐ c) Traversing
 - ☐ d) Insertion at middle ✓
- Which of the following is immutable?
 - ☐ a) String ✓
 - ☐ b) ArrayList
 - ☐ c) LinkedList
 - ☐ d) Queue
- How do you find duplicates in an array?
 - ☐ a) HashMap ✓

- b) Stack
 - c) Queue
 - d) Graph
 - 10. Which is faster: array or linked list (for access by index)?
 - a) Array ✓
 - b) Linked List
 - c) Both
 - d) None
 - 11. What is the default value of elements in an int array in Java?
 - a) -1
 - b) 1
 - c) 0 ✓
 - d) undefined
 - 12. What is the index of first element in an array?
 - a) 0 ✓
 - b) 1
 - c) -1
 - d) Depends on language
 - 13. Which method is used to reverse a string in Java?
 - a) reverse()
 - b) reverseString()
 - c) StringBuilder.reverse() ✓
 - d) strrev()
 - 14. What data structure would you use for implementing a leaderboard?
 - a) ArrayList
 - b) Priority Queue ✓
 - c) HashMap
 - d) Stack
 - 15. Which of the following is best for random access?
 - a) Array ✓
 - b) Linked List
 - c) Stack
 - d) Queue
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Linked Lists (10 MCQs)

16. Time complexity to insert at head of a singly linked list?
- a) $O(n)$
 - b) $O(1)$ ✓
 - c) $O(\log n)$
 - d) $O(n \log n)$
17. In a doubly linked list, each node has:
- a) 1 pointer

- b) 2 pointers ✓
- c) 3 pointers
- d) 0 pointers

18. Circular linked list is:

- a) Last node points to null
- b) Last node points to head ✓
- c) Head points to tail
- d) None

19. How to detect loop in linked list?

- a) DFS
- b) Two pointers (slow/fast) ✓
- c) BFS
- d) Sort

20. Which operation is easiest in singly linked list?

- a) Insertion at head ✓
- b) Deletion at tail
- c) Random access
- d) Sorting

21. What is the time complexity to delete a node at position n in a singly linked list?

- a) $O(1)$
- b) $O(n)$ ✓
- c) $O(n \log n)$
- d) $O(\log n)$

22. In linked list, memory is:

- a) Contiguous
- b) Non-contiguous ✓
- c) Fixed
- d) None

23. Sentinel node is used:

- a) For recursion
- b) For circular list
- c) To simplify operations ✓
- d) To reduce memory

24. Which is better for memory efficiency?

- a) Array

- b) Linked List ✓
- c) Queue
- d) Tree

25. Advantage of linked list over array?

- a) Fixed size
 - b) Better cache
 - c) Dynamic size ✓
 - d) Faster indexing
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Stacks and Queues (10 MCQs)

26. Which data structure uses LIFO order?

- a) Queue
- b) Stack ✓
- c) Array
- d) Linked List

27. Which of the following uses FIFO order?

- a) Stack
- b) Queue ✓
- c) Tree
- d) Graph

28. What is the time complexity of push operation in stack?

- a) $O(1)$ ✓
- b) $O(n)$
- c) $O(\log n)$
- d) $O(n \log n)$

29. Which method is used to insert an element in a queue?

- a) pop()
- b) push()
- c) enqueue() ✓
- d) delete()

30. Which method removes element from front of queue?

- a) dequeue() ✓
- b) push()
- c) pop()

- d) insert()

31. What data structure is used in recursion?

- a) Queue
- b) Stack ✓
- c) Tree
- d) Graph

32. Which data structure can be used to check balanced parentheses?

- a) Array
- b) Stack ✓
- c) Queue
- d) Tree

33. What happens when stack overflows?

- a) Program crashes ✓
- b) Data is lost
- c) Returns 0
- d) Starts again

34. What is the time complexity of dequeue operation in queue?

- a) $O(1)$ ✓
- b) $O(n)$
- c) $O(\log n)$
- d) $O(n \log n)$

35. Which is used to implement stack?

- a) Array ✓
- b) Graph
- c) Tree
- d) HashMap

Trees (10 MCQs)

36. Which tree traversal is used for expression trees?

- a) Inorder
- b) Postorder ✓
- c) Preorder
- d) Level Order

37. What is the maximum number of nodes in a binary tree of height h ?

- a) 2^h
- b) $2^{(h+1)} - 1$ ✓
- c) h^2
- d) $h * \log h$

38. In a BST, what is the time complexity for search in average case?

- a) $O(1)$
- b) $O(n)$
- c) $O(\log n)$ ✓
- d) $O(n \log n)$

39. Which traversal gives nodes in sorted order in BST?

- a) Preorder
- b) Inorder ✓
- c) Postorder
- d) Level Order

40. Which data structure is used in level order traversal?

- a) Stack
- b) Queue ✓
- c) Array
- d) Graph

41. What is a complete binary tree?

- a) Every level except last is full ✓
- b) All nodes have 2 children
- c) Every node has 1 child
- d) Tree with only leaf nodes

42. What is height of a tree with only root node?

- a) 0 ✓
- b) 1
- c) -1
- d) 2

43. AVL Tree is a type of:

- a) Binary Tree
- b) Balanced Binary Search Tree ✓
- c) Heap
- d) Graph

44. Which of these is not a tree traversal?

- a) Inorder
- b) BFS ✓
- c) Preorder
- d) Postorder

45. What is the number of null links in a binary tree with n nodes?

- a) n
- b) $n+1$ ✓
- c) $2n$
- d) $2n+1$

Graphs (10 MCQs)

46. Which of the following is used in DFS traversal?

- a) Queue
- b) Stack ✓
- c) Priority Queue
- d) Hash Table

47. Which algorithm works only on Directed Acyclic Graphs (DAG)?

- a) Kruskal
- b) Dijkstra
- c) Topological Sort ✓
- d) Prim

48. Which traversal is used in shortest path (unweighted graph)?

- a) DFS
- b) BFS ✓
- c) Inorder
- d) Preorder

49. What is the time complexity of DFS in adjacency list?

- a) $O(V+E)$ ✓
- b) $O(V^2)$
- c) $O(E^2)$
- d) $O(\log V)$

50. Which algorithm is used to detect cycle in directed graph?

- a) DFS ✓
- b) BFS

- c) Prim
- d) Dijkstra

51. What is a strongly connected graph?

- a) All nodes connected in one direction
- b) There is a path from every node to every other node ✓
- c) All nodes have even degree
- d) None

52. Which algorithm is used in Minimum Spanning Tree?

- a) DFS
- b) Prim's ✓
- c) BFS
- d) Bellman-Ford

53. Which graph representation is most space efficient for sparse graphs?

- a) Adjacency Matrix
- b) Adjacency List ✓
- c) Incidence Matrix
- d) Edge List

54. In a complete graph with n vertices, how many edges exist?

- a) n
- b) n^2
- c) $n(n-1)/2$ ✓
- d) $n(n+1)/2$

55. Which algorithm is used for shortest path with negative weights?

- a) Dijkstra
- b) Bellman-Ford ✓
- c) Prim
- d) Kruskal

Searching & Sorting (10 MCQs)

56. What is the time complexity of binary search?

- a) $O(n)$
- b) $O(\log n)$ ✓
- c) $O(1)$
- d) $O(n \log n)$

57. Merge Sort follows which algorithmic paradigm?

- a) Greedy
- b) Divide and Conquer ✓
- c) Dynamic Programming
- d) Backtracking

58. Which sorting algorithm is best for small datasets?

- a) Merge Sort
- b) Quick Sort
- c) Insertion Sort ✓
- d) Radix Sort

59. Which sorting algorithm is not comparison based?

- a) Merge Sort
- b) Radix Sort ✓
- c) Heap Sort
- d) Quick Sort

60. Quick Sort's worst-case time complexity is:

- a) $O(n \log n)$
- b) $O(\log n)$
- c) $O(n^2)$ ✓
- d) $O(n)$

61. Heap Sort uses which data structure?

- a) Stack
- b) Heap ✓
- c) Queue
- d) Tree

62. What is the best case of Bubble Sort?

- a) $O(n)$ ✓
- b) $O(n \log n)$
- c) $O(n^2)$
- d) $O(\log n)$

63. Which of the following is stable sort?

- a) Quick Sort
- b) Heap Sort
- c) Merge Sort ✓
- d) Selection Sort

64. What is the average case of Quick Sort?

- a) $O(n^2)$
- b) $O(n \log n)$ ✓
- c) $O(n)$
- d) $O(\log n)$

65. Which sorting algorithm is best for nearly sorted array?

- a) Merge Sort
- b) Quick Sort
- c) Insertion Sort ✓
- d) Selection Sort

Recursion & Backtracking (10 MCQs)

66. Which of the following problems is solved using backtracking?

- a) Tower of Hanoi
- b) N-Queens ✓
- c) Fibonacci
- d) Binary Search

67. What is the base case in recursion?

- a) The case with the smallest input ✓
- b) The infinite loop
- c) The recursive call
- d) None

68. Which data structure is used in recursion?

- a) Queue
- b) Stack ✓
- c) Linked List
- d) Heap

69. Which is NOT a characteristic of recursion?

- a) Base case
- b) Recursive call
- c) Iterative loop ✓
- d) Stack usage

70. Backtracking algorithm is:

- a) Greedy
- b) Brute-force with optimization ✓
- c) Divide and Conquer

- d) Dynamic Programming

71. What is the time complexity of recursive Fibonacci?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(2^n)$ ✓

72. Which problem is solved using recursion naturally?

- a) Tree Traversal ✓
- b) Searching
- c) Sorting
- d) Hashing

73. What happens if base case is missing in recursion?

- a) Returns null
- b) Infinite recursion ✓
- c) Compilation error
- d) Stack is unused

74. In recursion, every call is stored in:

- a) Heap
- b) Stack ✓
- c) Queue
- d) Array

75. The function calls itself in:

- a) Iteration
- b) Recursion ✓
- c) Inheritance
- d) Encapsulation

Dynamic Programming (10 MCQs)

76. What is dynamic programming?

- a) Divide & Conquer
- b) Solving problems by combining solutions to subproblems ✓
- c) Sorting method
- d) Recursion without memoization

77. Which of these is a classic DP problem?

- a) Prim's Algorithm
- b) Longest Common Subsequence ✓
- c) Binary Search
- d) DFS

78. In DP, overlapping subproblems mean:

- a) Subproblems are unrelated
- b) Same subproblems are solved multiple times ✓
- c) Problem can't be solved
- d) Using recursion

79. What is memoization?

- a) Looping
- b) Storing results of expensive function calls ✓
- c) Searching algorithm
- d) Sorting algorithm

80. Which method is bottom-up DP?

- a) Recursion
- b) Tabulation ✓
- c) Memoization
- d) Brute force

81. Knapsack problem is solved by:

- a) Greedy
- b) DP ✓
- c) BFS
- d) DFS

82. What is the space complexity of bottom-up DP?

- a) Always $O(1)$
- b) Depends on problem ✓
- c) $O(\log n)$
- d) $O(1)$

83. What is optimal substructure?

- a) Problem can be broken into optimal subproblems ✓
- b) Problem cannot be divided
- c) Only works in sorting
- d) Related to hash table

84. Which of these uses DP?

- a) N-Queens
- b) Matrix Chain Multiplication ✓
- c) Kruskal
- d) Prim's

85. Top-down approach is also called:

- a) Tabulation
- b) Memoization ✓
- c) Brute Force
- d) DFS



Greedy Algorithms (5 MCQs)

86. Greedy algorithms always:

- a) Choose local optimum ✓
- b) Choose global optimum
- c) Use recursion
- d) None

87. Greedy strategy fails in:

- a) Fractional Knapsack
- b) 0-1 Knapsack ✓
- c) Activity Selection
- d) Huffman Encoding

88. Huffman coding is based on:

- a) Sorting
- b) Greedy ✓
- c) Backtracking
- d) DFS

89. Which of the following uses greedy?

- a) Dijkstra ✓
- b) Bellman-Ford
- c) Binary Search
- d) Merge Sort

90. Activity Selection Problem uses:

- a) Dynamic Programming
- b) Greedy Approach ✓

- c) Divide & Conquer
 - d) Brute Force
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Complexity & Big-O (10 MCQs)

91. What is the time complexity of linear search?

- a) $O(1)$
- b) $O(n)$ ✓
- c) $O(\log n)$
- d) $O(n^2)$

92. Best case time complexity of binary search?

- a) $O(\log n)$
- b) $O(n)$
- c) $O(1)$ ✓
- d) $O(n \log n)$

93. Time complexity of accessing array element by index?

- a) $O(\log n)$
- b) $O(1)$ ✓
- c) $O(n)$
- d) $O(n \log n)$

94. What does Big O represent?

- a) Best case
- b) Average case
- c) Worst case ✓
- d) No case

95. Which of the following grows fastest?

- a) $O(1)$
- b) $O(n)$
- c) $O(n \log n)$
- d) $O(n!)$ ✓

96. What is the space complexity of recursive factorial?

- a) $O(n)$ ✓
- b) $O(\log n)$
- c) $O(1)$
- d) $O(n^2)$

97. Which is slower in growth rate?

- a) $O(n)$
- b) $O(n \log n)$ ✓
- c) $O(n^2)$
- d) $O(2^n)$

98. What is the time complexity of inserting in a hash table (average case)?

- a) $O(\log n)$
- b) $O(n)$
- c) $O(1)$ ✓
- d) $O(n^2)$

99. What is the time complexity of traversing a linked list?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$ ✓
- d) $O(n^2)$

100. Time complexity of enqueue operation in queue?

- a) $O(1)$ ✓
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n \log n)$